Reg. No. | | | | | | | | | | | [14AE33]

THIRD SEMESTER B.TECH EXAMINATION, DECEMBER – 2015

AEROSPACE VEHICLES AND SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Draw neat sketches wherever necessary.
3. All questions carry equal marks.

PART - A

1.	**a.**	Write brief notes on the contributions of Leonardo Da Vinci, Montgolfer Brothers and Sir George Cayley in the field of aviation.	**07**
	b.	Explain five layers of atmospheres and their properties.	**07**
2.	**a.**	Describe the functions of major airplane components.	**07**
	b.	Discuss in details five stresses that they act on an aircraft and its components.	**07**
3.		List out the main categories of aerospace propulsion engines and explain in brief the operating principles of each category with brief reference to the advantages of each category. Draw necessary simple sketches are required.	**14**
4.	**a.**	Discuss in details any five properties of metallic materials used in aerospace vehicles.	**07**
	b.	Explain the functioning of flight controls in a helicopters in reference to: i. Forward movement ii. Vertical flight, and iii. Helicopter Heading	**07**

PART - B

5.	**a.**	Draw neat diagram of a basic hydraulic system. Explain the function of each components in the system with neat diagram.	**04**
	b.	Explain the following:	
		i. Inflight refuelling	**04**
		ii. Fuel jettisioning	**03**
		iii. External fuel tanks	**03**
6.	**a.**	Explain in details the power assisted controls in an aircraft.	**07**
	b.	Explain the functioning of the major components of an aircrafts landing gear.	**07**

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7.	**a.**	State and explain the Kepler's laws of space dynamics.	**07**
	b.	Discuss the important orbits for satellites.	**07**
8.		Discuss the classification of rockets based on the propellants used. Explain any three types in detail with neat diagrams.	**14**